Originally when I got my hands on a 9835 diagnostic ROM and RAM card to reverse engineer, I built a clone of the RAM card and added to ROM image to my ROM emulator that is a card that is plugged into the memory bus and provided a jumper so I could disable the drag ROM image. That meant that I had to take the cover off to enable or disable the image and that for old pretty quick so then I set about creating a single card that plugs into a ROM slot in the front.

I could base the design on the diagram RAM card but needed to alter the address decoding. On the RAM card the decoding is done by a single 13 input NAND gate and three inverters to decode to block 0x25 and place the RAM at the top 256 words of the block. The ROM is in the same block but occupies the bottom 4K words. What I did was to separate the decoding of the block address and then have separate decoders for the RAM and ROM space and used the decoded block address to qualify them.

You will note on the schematic that the RAM caps I used are 16K parts, but I grounded the top 3 address lines so only 256 byte in each chip are actually used. I chose these because they are 8 bit parts in a skinny package so in the end saved space. The original RAM card used 256x4 SRAMs the part you use is not critical and if for some reason you want to enable more than 256 words you will need to adjust the address decoding for the RAM, but I do not think there is any point undoing that, it seems unlikely that the diagnostics will use any more that the 256 words, the diagnostics just need some scratch pad space out of the normal memory space to be able to fully test the machine.

I built a hand wired prototype that I use, but as usual I have not laid out a PCB, I do not currently have the skill to do that, I would like to learn but since I am pushing 70 I may never get there.

This package includes a schematic and a image of the ROM please note this is a 16 bit image and will need to be split to write to the two 3732 EPROMs.

Paul.